Limit of functions

F18CD Week 7

REVISION

1. Intervals
2. Definition of Functions
3. Domain, Codomain, Range
4. Functions' operations
5. Bounded functions

1. Intervals

An interval is closed if it contains each of its end points (if the end point exists) and open if it contains neither of its end points (if the end point exists).

We use a square bracket [ ] if the interval contains the corresponding end point and a round bracket ( ) if it does not.

Intervals (cont.)

Let $a, b \in \mathbb{R} \ni a < b$. Define the following sets (and notations)

Notation	Set	End point(s)	Open/closed interval?
$(a,\ b)$ or $]a,\ b[$	$\{x \in \mathbb{R}\colon\ a < x < b\}$	a and b	Open
$[a,\ b]$	$\{x \in \mathbb{R}\colon\ a \leq x \leq b\}$	a and b	Closed
$(a,\ b]$ or $]a,\ b]$	$\{x \in \mathbb{R}\colon\ a < x \leq b\}$	a and b	Neither/half open/ left-open/right-closed
$[a,\ b)$ or $[a,\ b[$	$\{x \in \mathbb{R}\colon\ a \leq x < b\}$	a and b	Neither/half open/ right-open/left-closed

Intervals (cont.)

(a, ∞)	$\{x \in \mathbb{R}: x > a\}$	a only	Open
$[a, \infty)$	$\{x \in \mathbb{R}: x \geq a\}$	a only	Closed/left-closed
$(-\infty, b)$	$\{x \in \mathbb{R}: x < b\}$	b only	Open
$(-\infty, b]$	$\{x \in \mathbb{R}: x \leq b\}$	b only	Closed/right-closed
$(-\infty,\infty)$	$\mathbb{R}$	Not exist	Both open and closed

2. Definition of functions

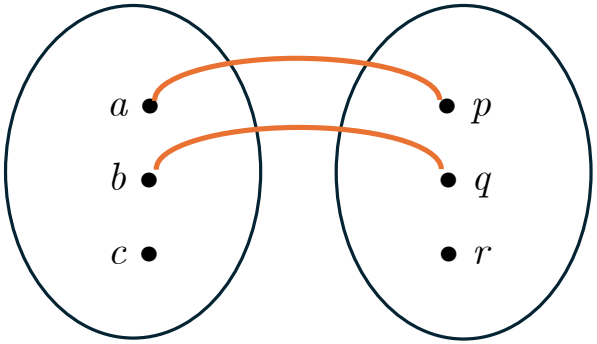
Definition 2.1

Let A be a set such that $A \subseteq \mathbb{R}$. A function f from A to $\mathbb{R}$ (often called a 'real-valued' function), written as $f : A \to \mathbb{R}$ is an assignment which map each element $a \in A$ to a unique number $f(a) \in \mathbb{R}$. A is called the domain of f.

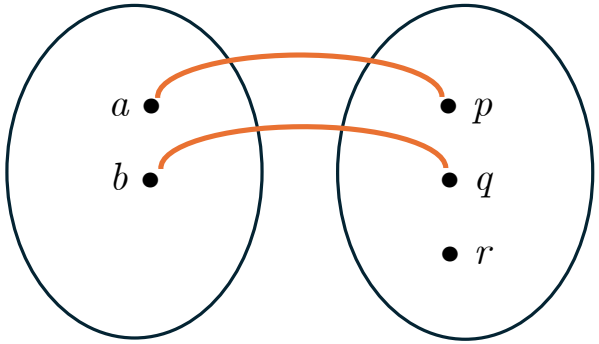
Which of the following mappings is a function?

Consider a mapping from $\{a, b, c\}$ to $\{p, q, r\}$.

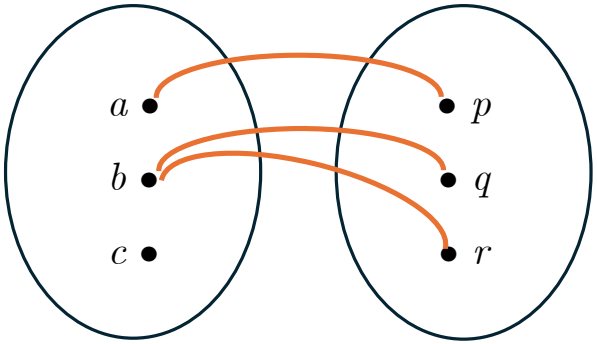
This is not a function.



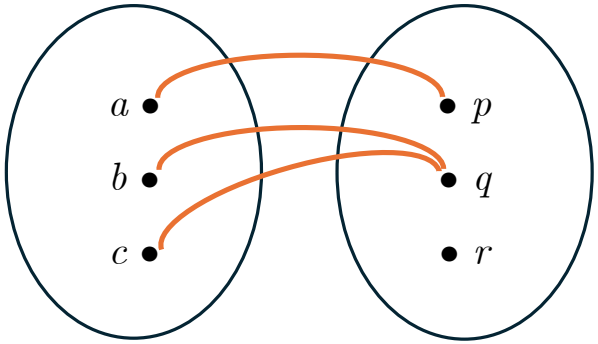
This is a function.



This is not a function.



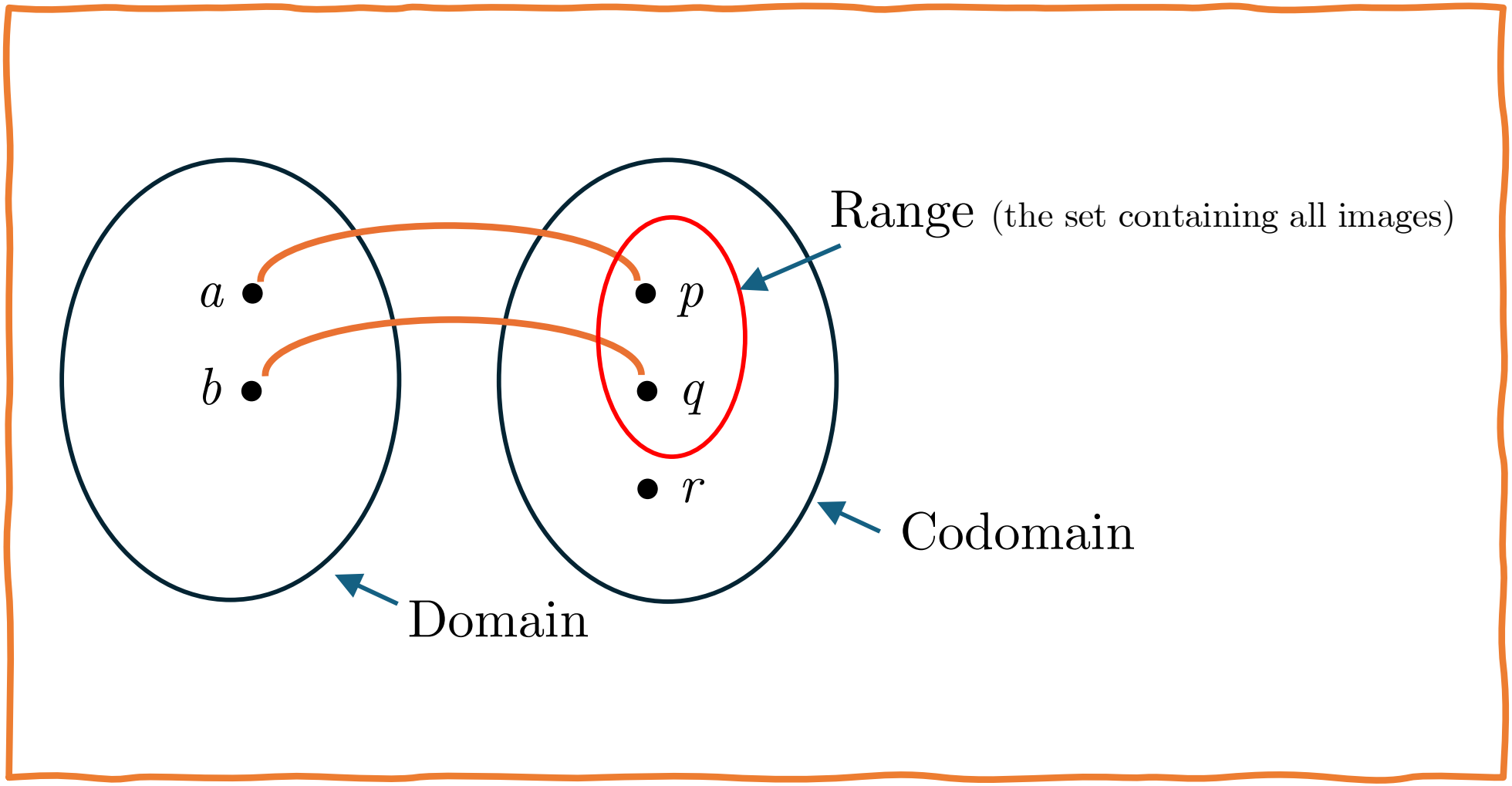
This is a function.



3. Domain, codomain and range

Let A and B be non-empty sets and let $f : A \to B$ be a function. Then we say domain of f is A and co-domain of f is B. The set A that we define must not be undefined for function f. The range of f is a set $C \subseteq B$, where $C = \{f(a) : \text{for all } a \in A\}$.

Remark: A real function is a function with its codomain is $\mathbb{R}$.


Range (the set containing all images)
a
b
p
q
r
Codomain
Domain

Example 2.3 (i) and (ii)

(i) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ and $g : (1, \infty) \to \mathbb{R}$ be given as

$$f(x) = \sin\left(\tfrac{1}{x}\right) \text{ and } g(x) = \sin\left(\tfrac{1}{x}\right).$$

Note that $f(x) \neq g(x)$ because they do not have the same domain.

(ii) Functions can be defined in piecewise. For example, a piecewise function $h : \mathbb{R} \to \mathbb{R}$ defined by

$$h(x) = \begin{cases} \sin x, & x \geq 0 \\ \cos x, & x < 0 \end{cases}.$$

Example 2.3 (iii)

(iii) Let $d : \mathbb{R} \to \mathbb{R}$ be a function given by

$$d(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}.$$

This function is called the Dirichlet function.

Thomae's function

$$t(x) = \begin{cases} 1, & x = 0 \\ \frac{1}{n}, & x = \frac{m}{n} \in \mathbb{Q} \setminus \{0\}, \gcd(m,n) = 1, n > 0 \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

4. Operations on functions

Let $f : S_1 \to \mathbb{R}$ and $g : S_2 \to \mathbb{R}$ be two functions and set $S = S_1 \cap S_2$.

Then the following are functions

$$\lambda f : S_1 \to \mathbb{R}$$
$$f + g, fg : S \to \mathbb{R}$$
$$\tfrac{f}{g} : S \setminus \{x : g(x) = 0\} \to \mathbb{R}$$
$$\left|f\right| : S_1 \to \mathbb{R}$$

defined by

$$\Big(\lambda f\Big)(x) = \lambda f(x) \qquad \forall x \in S_1$$
$$\Big(f + g\Big)(x) = f(x) + g(x) \qquad \forall x \in S$$
$$\Big(fg\Big)(x) = f(x)g(x) \qquad \forall x \in S$$
$$\tfrac{f}{g}(x) = \tfrac{f(x)}{g(x)} \quad \text{providing } g(x) \neq 0, \forall x \in S$$
$$\Big|f\Big|(x) = \Big|f(x)\Big|$$

Example

Suppose $f : \mathbb{R}^+ \cup \{0\} \to \mathbb{R}$ such that $f(x) = \sqrt{x}$ and $g : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ such that $g(x) = \dfrac{1}{x}$. Then we could define some new functions $p(x), q(x), r(x) : \mathbb{R}^+ \to \mathbb{R}$

$$p(x) := f(x) + g(x) = \sqrt{x} + \frac{1}{x}$$
$$q(x) := f(x) - g(x) =$$
$$r(x) := f(x)g(x) =$$

Example

Suppose $f, g : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^2$ and $g(x) = x$.

Let $h(x) : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ such that $h(x) := \dfrac{f(x)}{g(x)} = \dfrac{x^2}{x} = x$.

What is the difference between a function $g(x) = x$ and $h(x) = \dfrac{x^2}{x}$?

Are they the same?

Question

Suppose $f, g : \mathbb{R} \to \mathbb{R}$ such that $f(x) = x^2 - 1$ and $g(x) = x + 1$.

What is $h(x) := \dfrac{f(x)}{g(x)}$?

Composite functions

Let A, B and C be nonempty sets.

Suppose $f : A \to B$ and $g : B \to C$ be functions.

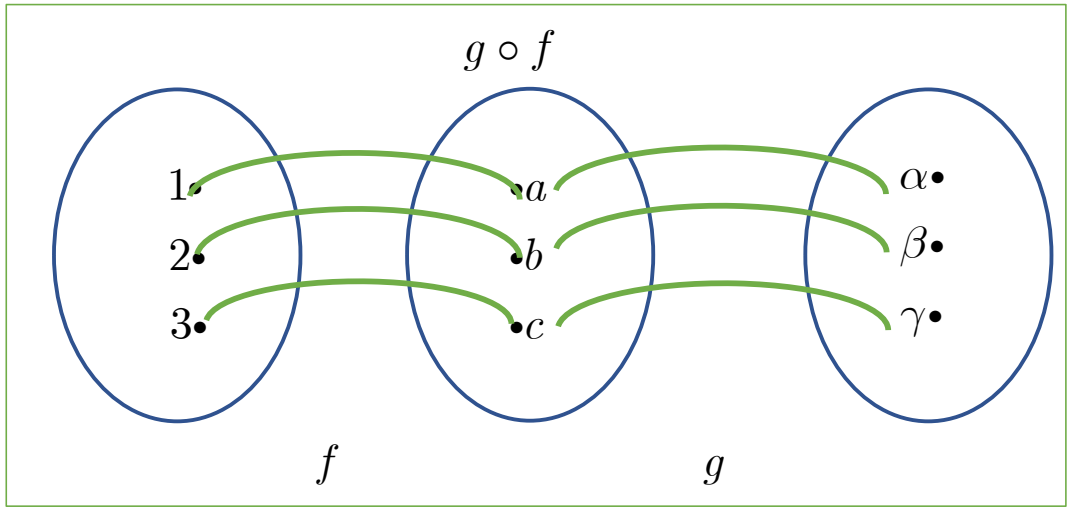
Then we define the composition of f and g, written as

$$g \circ f \,,$$

is defined as

$$g \circ f = g\Big(f(x)\Big).$$

where $h = g \circ f$.



Domain of composite function

Write $D_g =$ domain of g, $D_f =$ domain of f. Then

$$D_{g \circ f} = \left\{ x \in D_f : f(x) \in D_g \right\}.$$

Example

Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R}^+ \cup \{0\} \to \mathbb{R}$ such that $f(x) = x^2$ and $g(x) = \sqrt{x}$. Then

$$f \circ g = f(g(x)) = f(\sqrt{x}) = \left(\sqrt{x}\right)^2 = x$$

$$g \circ f = g(f(x)) = g(x^2) = \sqrt{x^2} = \left|x\right|$$

$$D_{f\circ g} = \left\{x \in \mathbb{R}^+ \cup \{0\} : \sqrt{x} \in \mathbb{R}\right\} = \mathbb{R}^+ \cup \{0\}$$

$$D_{g\circ f} = \left\{x \in \mathbb{R} : x^2 \in \mathbb{R}^+ \cup \{0\}\right\} = \mathbb{R}$$

Example

Let $f : \mathbb{R}^+ \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \frac{1}{\sqrt{x}}$ and $g(x) = x + 1$.

Then

$$f \circ g = \frac{1}{\sqrt{x+1}}$$

$$g \circ f = \frac{1}{\sqrt{x}} + 1$$

$$f \circ f = \frac{1}{\sqrt{\frac{1}{\sqrt{x}}}} = x^{\frac{1}{4}}$$

$$D_{f\circ g} = \left\{x \in \mathbb{R} : x + 1 \in \mathbb{R}^+\right\} = (-1, \infty)$$

$$D_{g\circ f} = \left\{x \in \mathbb{R}^+ : \frac{1}{\sqrt{x}} \in \mathbb{R}\right\} = \mathbb{R}^+.$$

$$D_{f\circ f} = \left\{x \in \mathbb{R}^+ : \frac{1}{\sqrt{x}} \in \mathbb{R}^+\right\} = \mathbb{R}^+.$$

5. Bounded above, bounded below, bounded & unbounded

Definition 2.8.

Let $D \subseteq \mathbb{R}$ be a set and let $f : D \to \mathbb{R}$ be a function. Then we say f is

i. bounded above if there exists $U \in \mathbb{R}$ such that $f(x) \leq U \ \forall x \in D$,

ii. bounded below if there exists $L \in \mathbb{R}$ such that $f(x) \geq L \ \forall x \in D$,

iii. bounded if f is bounded above and bounded below,

iv. unbounded if f is not bounded.

Example 2.9

(a) The function $f : [1, \infty) \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is bounded on $[1, \infty)$, i.e., f is bounded below by 0 and it is bounded above by 1.

However, $g : (0, 1] \to \mathbb{R}$ defined by $g(x) = \frac{1}{x}$ is unbounded.

(b) The function $h : [0, 3] \to \mathbb{R}$ defined by $h(x) = x^2$ is bounded on $[0, 3]$, i.e., h is bounded below by 0 and it is bounded above by 9.

Remark: The domain decides whether a function is bounded or not.

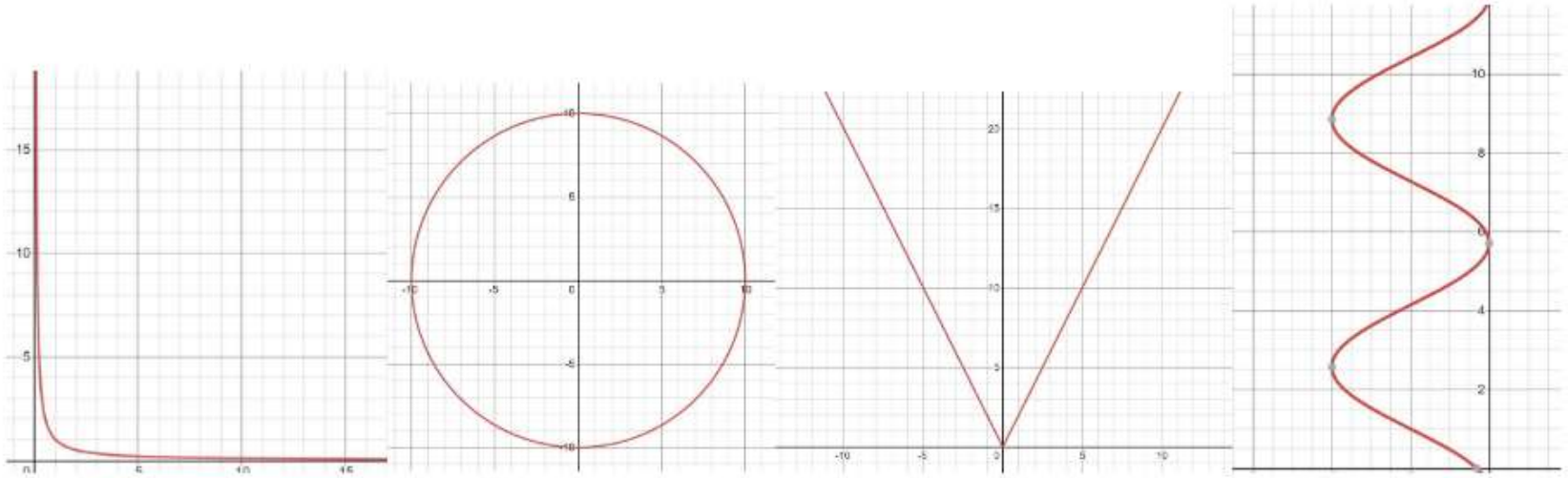
Theorem 2.10

A function $f : D \to \mathbb{R}$ is bounded if and only if there exist a real number $K \geq 0$ such that

$$\left|f(x)\right| \leq K \quad \forall x \in D\,.$$

The signs $<$ and $\leq$ are interchangeable

$$y = \frac{1}{x} \qquad x^2 + y^2 = 10^2 \qquad y = |2x| \qquad \sin(y-1) = -\frac{1}{2}x - 1$$



Determine which of the above are functions? Which function is bounded?

$y = -\big|\,|\,2x\,| - 4\big| + 4$

$y = -\big|3(x-2)\big| + 5$

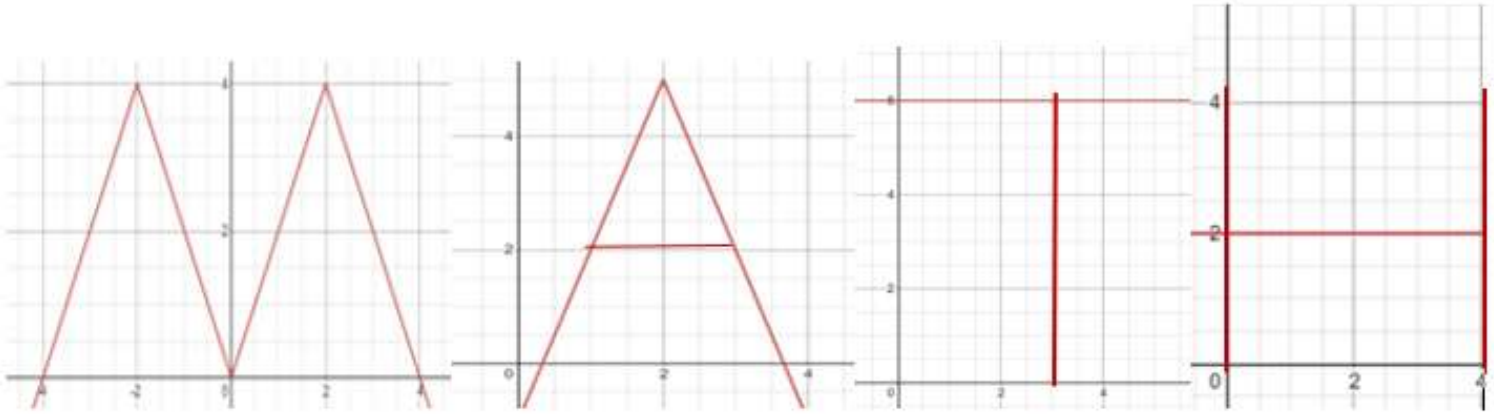
$y = 2,\ \frac{3}{2} \le x \le \frac{5}{2}$

$y = 4,\ 0 \le x \le 5$

$x = 3,\ 0 \le y \le 5$

$y = 2,\ 0 \le x \le 4$

$x = 0, x = 4,\ 0 \le y \le 4$



Determine which of the above are functions? Which function is bounded?

6. Limit of $f(x)$ as x tends to ∞ (or $-\infty$)

Recap limit of a convergence sequence

Recall definition of the limit of a sequence when $n \to \infty$. For all $\varepsilon > 0$

$$\exists N \in \mathbb{N} \ni n > N \Rightarrow \left|a_n - L\right| < \varepsilon \ \forall \ n \in \mathbb{N}.$$

We write $\lim_{n\to\infty} a_n = L$.

We imitate the definition above and extend the definition to limit of a function $f(x)$ when $x \to \infty$. A function can be seen as a sequence but the input is not just positive integers, but (usually) real numbers.



Definition 6.1. $\lim_{x \to \infty} f(x) = L$

Let $f : \left(a, \infty\right) \to \mathbb{R}$. Then we say that f converges to a limit L as $x \to \infty$, if,

$$\forall \varepsilon > 0 \ \exists X \in \left(a, \infty\right) \ni \left|f(x) - L\right| < \varepsilon \ \forall x > X \, .$$

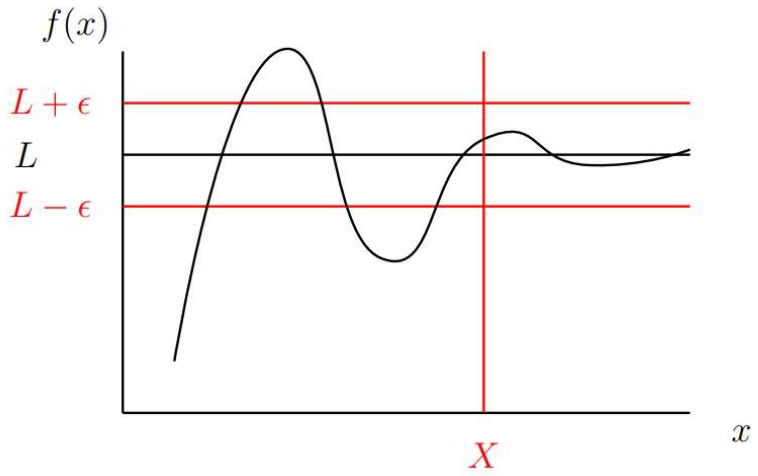
We write $\lim_{x \to \infty} f(x) = L$ (or $f(x) \to L$ as $x \to \infty$).

Comparison between limit of sequence and function

$$a : \mathbb{N} \to \mathbb{R} \qquad \lim(a_n) = L \Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|a_n - L\right| < \varepsilon \qquad \forall\ n > N.$$

$$f : (a, \infty) \to \mathbb{R} \qquad \lim_{x \to \infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists X > a \ni \left|f(x) - L\right| < \varepsilon \qquad \forall\ x > X.$$

or $X \in (a, \infty)$



In other words, $f(x) \longrightarrow L \quad \forall\ x > X$.

Example. Prove from the definition that $\lim_{x\to\infty} \frac{x}{1+x} = 1$.

Suppose $a, b \in \mathbb{R}^+ \ni b > a$.

Then $\exists n \in \mathbb{N} \ni na > b$ (Archimedean property).

Let $\varepsilon = \frac{1}{b} > 0$ and write $X = na > 0$.

Then $na > b \Rightarrow X > \frac{1}{\varepsilon}$. Thus, for $x > X > 0$,

$$\left|\frac{x}{1+x} - 1\right| = \left|\frac{1}{1+x}\right| < \frac{1}{x} < \frac{1}{X} < \varepsilon .$$

Definition 6.1. $\lim_{x \to -\infty} f(x) = L$

Let $f : \left(-\infty, b\right) \to \mathbb{R}$. Then we say that f converges to a limit L as $x \to -\infty$, if,

$$\forall \varepsilon > 0 \ \exists X \in \left(-\infty, b\right) \ni \left|f(x) - L\right| < \varepsilon \ \forall \ x < X.$$

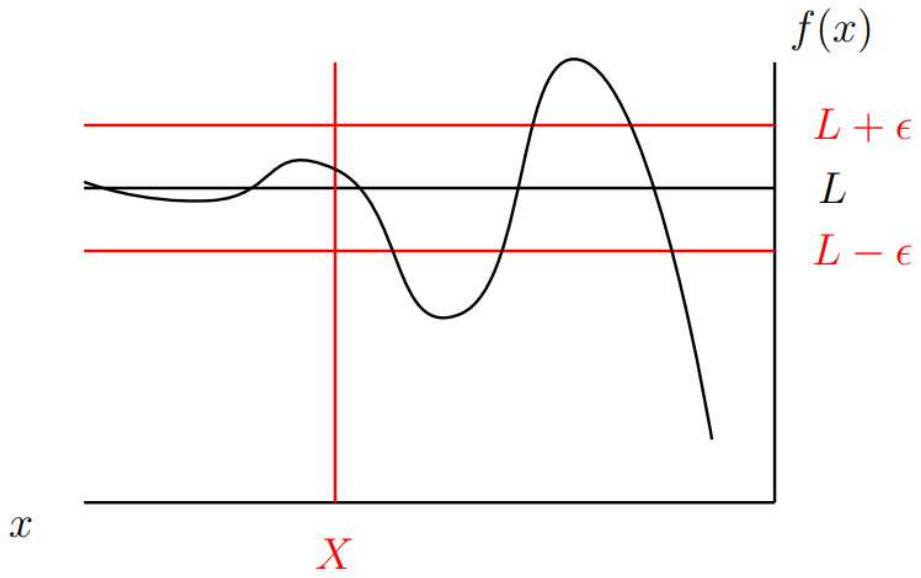
We write $\lim_{x \to -\infty} f(x) = L$ (or $f(x) \to L$ as $x \to -\infty$).

$$a : \mathbb{N} \to \mathbb{R} \qquad \lim(a_n) = L \Leftrightarrow \forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|a_n - L\right| < \varepsilon \qquad \forall\ n > N.$$

$$f : (a, \infty) \to \mathbb{R} \qquad \lim_{x \to \infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists X > a \ni \left|f(x) - L\right| < \varepsilon \qquad \forall\ x > X.$$

$$f : (-\infty, b) \to \mathbb{R} \qquad \lim_{x \to -\infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists X < b \ni \left|f(x) - L\right| < \varepsilon \qquad \forall\ x < X.$$

or $(-\infty,\ b)$



In other words, $f(x) \to L \quad \forall \quad x < X$.

Example

Example. Prove from the definition that $\lim_{x\to-\infty} \frac{x}{1+x} = 1$.

Let $a,b \in \mathbb{R}^+ \ni b > a$. Then $\exists n \in \mathbb{N} \ni na > b$.

Let $\varepsilon = \frac{1}{b} > 0$ and $-X = na > 0$. Then $-X > \frac{1}{\varepsilon} > 0$.

Now, $x < X < 0 \Rightarrow -x > -X > 0 \Rightarrow -\frac{1}{x} < -\frac{1}{X}$.

So $\left|\frac{x}{1+x} - 1\right| = \left|\frac{1}{1+x}\right| < \frac{1}{|x|} = \frac{1}{-x} < \frac{1}{-X} < \varepsilon$.

Definition (f(x) diverges to infinity when x tends to infinity)

Let $f : \left(a, \infty\right) \to \mathbb{R}$. Then we say that f diverges to ∞ , if,

$$\forall M > 0 \ \exists X \in \left(a, \infty\right) \ni f(x) > M \ \forall x > X \, .$$

We write $\lim\limits_{x \to \infty} f(x) = \infty$.

Comparison between limit in sequence and limit in function

$a : \mathbb{N} \to \mathbb{R}$	$\lim(a_n) = \infty \Leftrightarrow \forall M > 0, \exists N \in \mathbb{N} \ni a_n > M$	$\forall\ n > N.$
$f : (a, \infty) \to \mathbb{R}$	$\lim_{x \to \infty} f(x) = \infty \Leftrightarrow \forall M > 0, \exists X > a \ni f(x) > M$	$\forall\ x > X.$

Example

Example. Prove from definition that $\lim_{x\to\infty}\left(x^3 - x^2\right) = \infty$.

Proof. Let $M \in \mathbb{R}^+$. Choose $X \geq \max\left\{2, \sqrt{M}\right\}$.

Then $\forall x > X$

$$x^3 - x^2 = x^2(x-1) > X^2(X-1) > M \times 1 = M\,.$$

Note that by choosing $\max\left\{2, \sqrt{M}\right\}$, we guarantee $X \geq 2$ and $X \geq \sqrt{M}$.

$$X \geq \max\left\{2, \sqrt{M}\right\}.$$

Case 1. If $\sqrt{M} \geq 2$. Then $X^2(X-1) \geq M\left(\sqrt{M}-1\right) \geq M\left(2-1\right) = M$.

Case 2. If $2 \geq \sqrt{M}$. Then $X^2(X-1) \geq 4(2-1) = 4 \geq M$.

Try to define the following limits

$$\lim_{x \to \infty} f(x) = -\infty$$

$$\lim_{x \to -\infty} f(x) = \infty$$

$$\lim_{x \to -\infty} f(x) = -\infty$$

Definitions

$a : \mathbb{N} \to \mathbb{R}$ $\quad \lim(a_n) = \infty \Leftrightarrow \forall M > 0, \exists N \in \mathbb{N} \ni a_n > M \quad \forall\ n > N.$

$f : (a, \infty) \to \mathbb{R}$ $\quad \lim_{x \to \infty} f(x) = \infty \Leftrightarrow \forall M > 0, \exists X > a \ni f(x) > M \quad \forall\ x > X.$

$f : (-\infty, b) \to \mathbb{R}$ $\quad \lim_{x \to -\infty} f(x) = \infty \Leftrightarrow \forall M > 0, \exists X < b \ni f(x) > M \quad \forall\ x < X.$

$f : (a, \infty) \to \mathbb{R}$ $\quad \lim_{x \to \infty} f(x) = -\infty \Leftrightarrow \forall M < 0, \exists X > a \ni f(x) < M \quad \forall\ x > X.$

$f : (-\infty, b) \to \mathbb{R}$ $\quad \lim_{x \to -\infty} f(x) = -\infty \Leftrightarrow \forall M < 0, \exists X < b \ni f(x) < M \quad \forall\ x < X.$

7. Limit of $f(x)$ as x tends to a point.

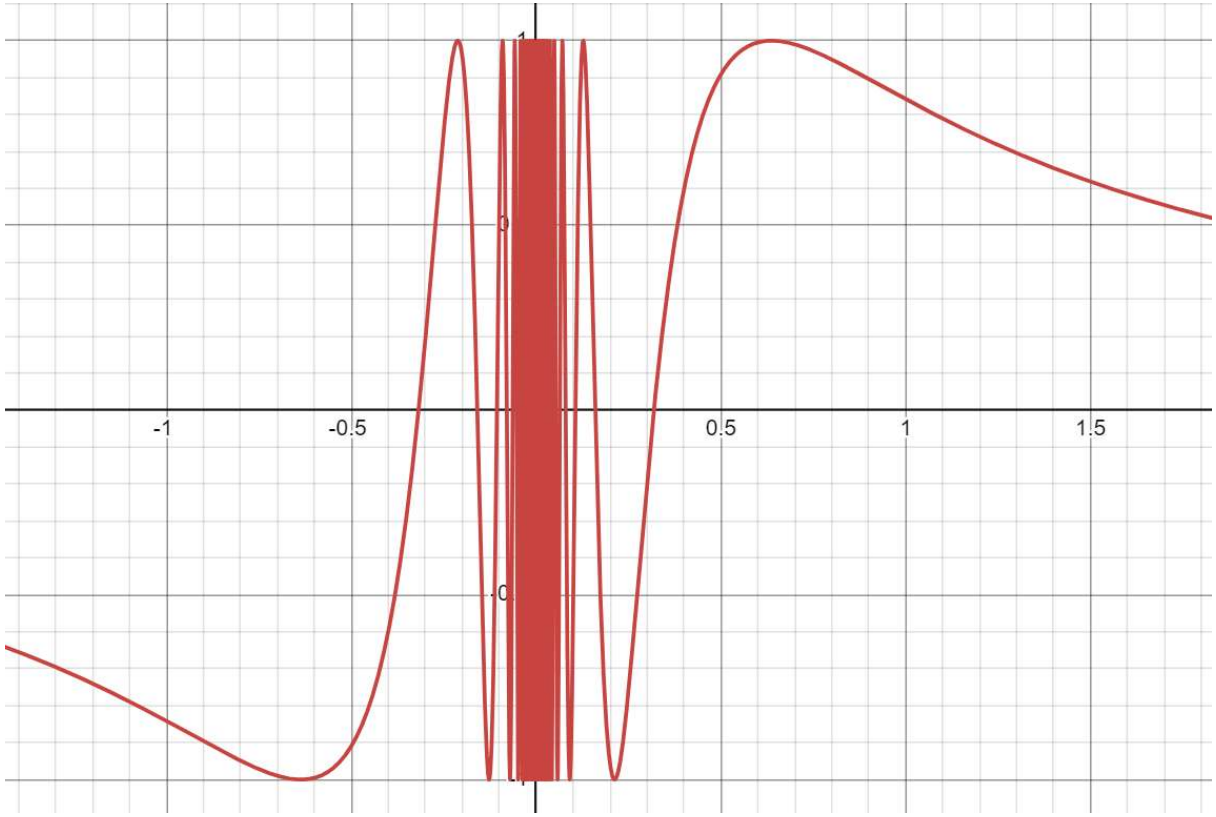
Motivation (from Example 6.3)

- Finding a limit at a point is not substitution of a value of x to obtain the answer.
- We provide the next few examples for you to study, what is the limit of these functions when $x \to 0$?

Let $f, g, h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be functions defined by

$$f(x) = \sin\left(\frac{1}{x}\right), \qquad g(x) = x \sin\left(\frac{1}{x}\right), \qquad h(x) = \frac{\sin x}{x}.$$

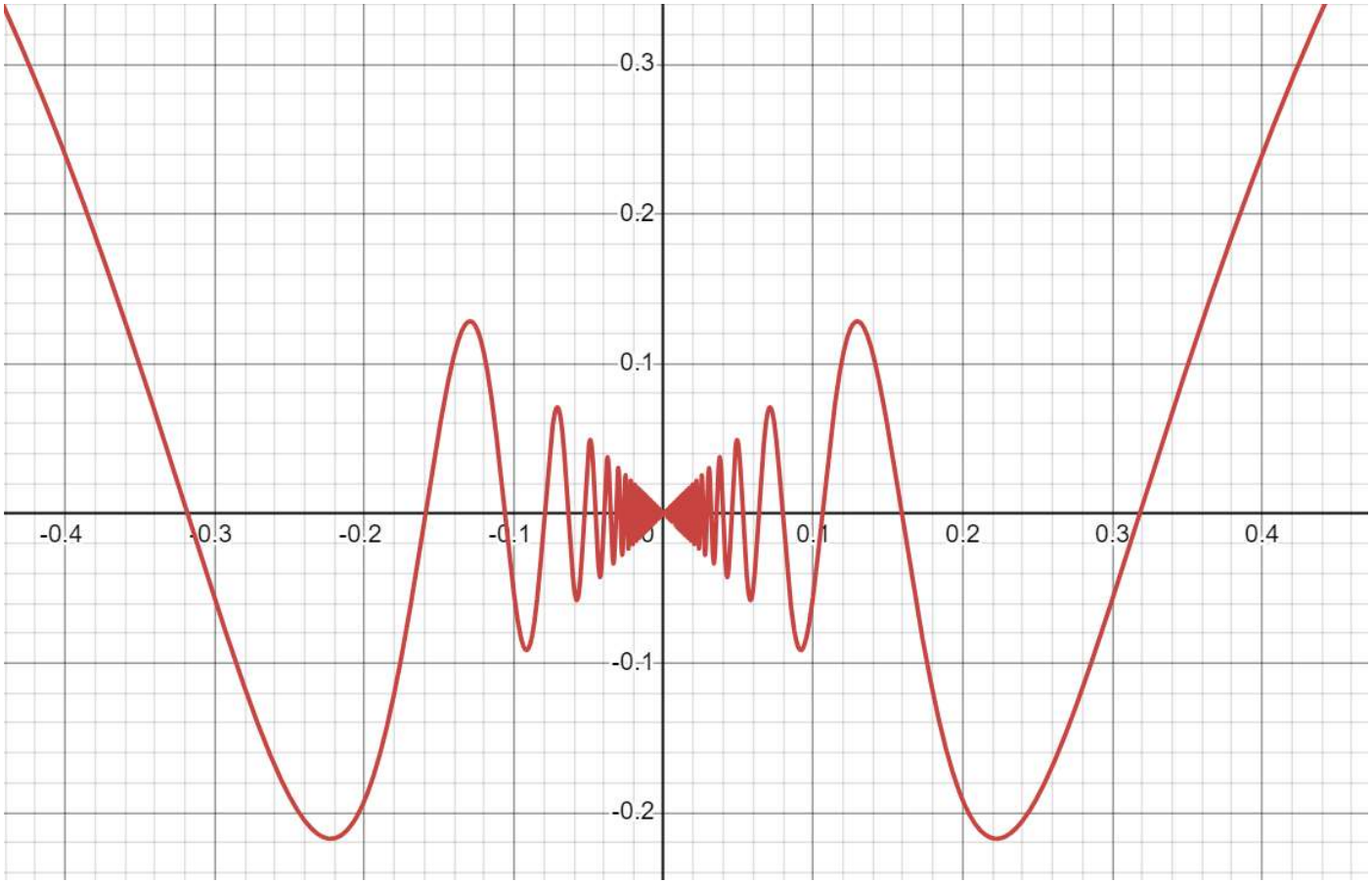
Graph of $f(x) = \sin\frac{1}{x}$



Our guess is....

$$\lim_{x \to 0} f(x) \not\exists$$

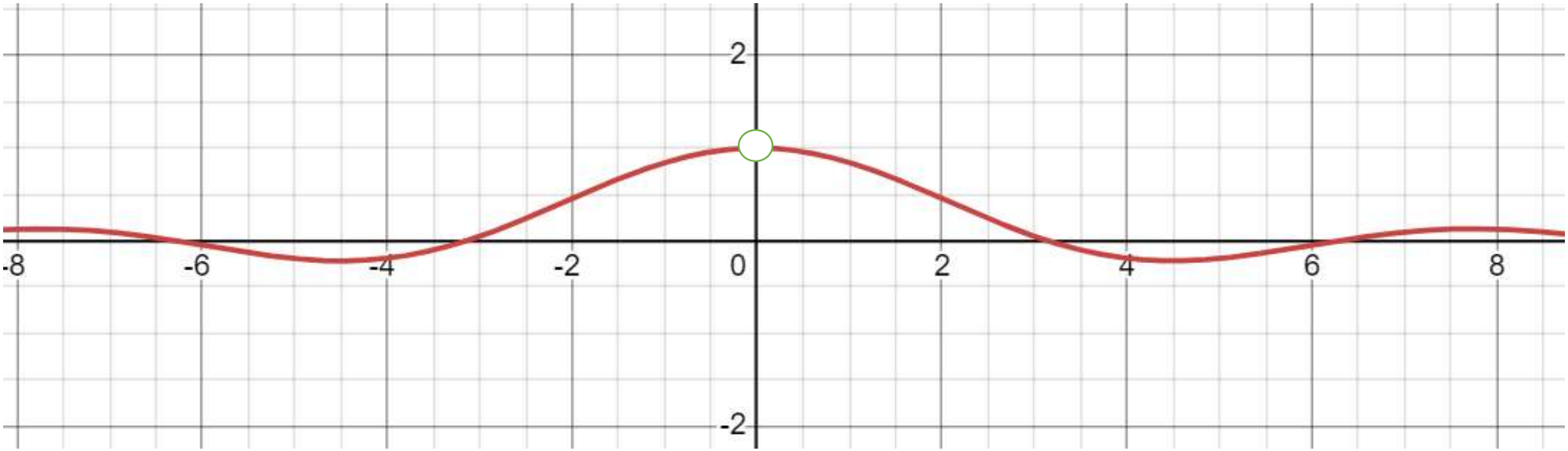
Graph of $g(x) = x\sin\frac{1}{x}$



Our guess is….

$$\lim_{x \to 0} g(x) = 0$$

Graph of $h(x) = \frac{\sin x}{x}$



Our guess is....

$$\lim_{x \to 0} h(x) = 1$$

Remark: We only look at values of $f(x)$ with $0 < |x - a|$, that is when $x \neq a$.

In other words, limit of $f(x)$ as $x \to a$

- DOES NOT need f to be defined at $x = a$
- DOES NOT depend on $f(a)$, even if $f(x)$ is defined at $x = a$

Definition 5.1. Limit points (boundary points)

A point $c \in \mathbb{R}$ is said to be a limit point of a set D if

$$\forall \varepsilon > 0 \ \exists x \in D \ni 0 < \left| x - c \right| < \varepsilon \, .$$

Definition 5.1. Closure of a set

The closure of the set D is the union of D and all its limit points, denoted by $\overline{D}$.

Remark: The closure of a set D is the smallest closed set containing D. The term 'closure' is also used to refer to a 'closed' version of a given set.

Example 5.2. Limit point

Show that 0 is a limit point of $D = \left(0, \infty\right)$.

Proof. $\forall \varepsilon > 0$, choose $x = \frac{\varepsilon}{2}$. Then $x \in D$ and

$$0 < \left|x - 0\right| < \varepsilon .$$

Thus, 0 is a limit point of $D = \left(0, \infty\right)$.

Example: Not a limit point

Show that –1 is not a limit point of $D = \left(0, \infty\right)$.

Proof. Suppose –1 is a limit point of $D = \left(0, \infty\right)$.

Let $\varepsilon = \frac{1}{2}$. Then $\exists x \in \mathbb{R}^+ \ni 0 < \left|x + 1\right| < \frac{1}{2}$.

Hence, $-\frac{3}{2} < x < -\frac{1}{2}, x \neq -1$, which mean $x \notin \mathbb{R}^+$,

a contradiction.

Example: Not a limit point

Show that any real number $a > 1$ is not a limit point of $D = (0,1)$.

Proof. Write $a = 1 + \delta$ for some $\delta > 0$. Suppose a is a limit point of D. Then $\exists x \in D$ such that $0 < |x - a| < \varepsilon$ for all $\varepsilon > 0$. Now, we choose ε such that $\varepsilon < \delta$. This is possible because $\mathbb{R}$ is dense. Then we have

$$1 < 1 + \delta - \varepsilon = a - \varepsilon < x < a + \varepsilon,$$ a contradiction.

Exercise. Prove that any real number $b < 0$ is not a limit point of $\big(0,1\big)$.

Remark

Let $f : D \to \mathbb{R}$ be a function.

Suppose $D = \big(a, b\big)$. Then $\bar{D} = \big[a, b\big]$.

But when we have $D = (a, \infty)$, then $\bar{D} = [a, \infty)$.

Theorem 7.1. Limit points

A point $c \in \mathbb{R}$ is a limit point of D if and only if $c = \lim\left(a_n\right)$ for some sequence $\left(a_n\right) \subseteq D$ with $a_n \neq c$ for all $n \in \mathbb{N}$.

Proof. $\Rightarrow$) By the definition of limit points, choose $x = a_n$,

where a_n are closed to c but not equal to c for all $n \in \mathbb{N}$

$$\forall \varepsilon > 0 \ \exists a_n \in D \ni 0 < \left| a_n - c \right| < \varepsilon \ \forall n \in \mathbb{N}.$$

This fits the definition of $c = \lim\left(a_n\right)$, where $a_n \in D \setminus \{c\}$.

$\Leftarrow$) If $\lim\left(a_n\right) = c$ with $a_n \in D \setminus \{c\}$, then by the definition of sequence,

$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ni n > N \Rightarrow \left|a_n - c\right| < \varepsilon \ \forall n \in \mathbb{N}$.

Now $a_n \neq c$, so we have $\left|a_n - c\right| > 0$ and hence c is a limit point of D.

Remark: It is important to remember that limit points of

D do not necessarily belong to the set D unless D is closed.

Definition 6.2. Limit of Function

Let $f : D \to \mathbb{R}$ be a function. Let $c \in \bar{D}$.

Then f converges to $L \in \mathbb{R}$ as x approaches c, if,

$\forall \varepsilon > 0 \ \exists \delta > 0$ such that $\left|f(x) - L\right| < \varepsilon$,

whenever $0 < \left|x - c\right| < \delta$, with $x \in D$.

We write: $\lim_{x \to c} f(x) = L$.

In fact, we are more interested on limit points!!

Let $f : D \to \mathbb{R}$ be a function. Let $c \in \bar{D}$ and $x \in D$.

$\boxed{\lim_{x \to c} f(x) = L}$ if and only if $\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } 0 < \left|x - c\right| < \delta \Rightarrow \left|f(x) - L\right| < \varepsilon}$

Example

Prove that $\lim_{x \to 2} 10x = 20$.

Proof.

$$0 < \left|x - 2\right| < \delta \Rightarrow \left|10x - 20\right| = 10\left|x - 2\right| < 10\delta .$$

Now if we choose $\delta = \frac{\varepsilon}{10}$. Then

$$0 < \left|x - 2\right| < \delta \Rightarrow \left|10x - 20\right| < 10\delta = \varepsilon .$$

Example

Prove that $\lim_{x\to 3} x^2 = 9$.

Proof. $0 < \left|x-3\right| < \delta_1 \Rightarrow \left|x+3\right| = \left|x-3+6\right| \le \left|x-3\right| + 6 < \delta_1 + 6$

$0 < \left|x-3\right| < \delta_2 \Rightarrow \left|x^2-9\right| = \left|(x-3)(x+3)\right| = \left|x-3\right|\left|x+3\right| < \delta_2\left|x+3\right|$

Now if we choose $\delta_1 = 1$ and $\delta_2 = \frac{\varepsilon}{7}$, and $\delta = \min\left\{\delta_1, \delta_2\right\} = \min\left\{1, \frac{\varepsilon}{7}\right\}$.

Then $0 < \left|x-3\right| < \delta \Rightarrow \left|x^2-9\right| < \delta_2(\delta_1+6) = \frac{\varepsilon}{7}(1+6) = \varepsilon$.

Why $\delta = \min\left\{\delta_1, \delta_2\right\}$?

Verify.... Note that

$$\left|x^2-9\right| = \left|x-3\right|\left|x-3+6\right| \le \left|x-3\right|\left(\left|x-3\right|+6\right)$$

Case 1: If $1 < \frac{\varepsilon}{7}$. Then $\min\{1, \frac{\varepsilon}{7}\} = 1$.

So, $0 < \left|x-3\right| < \delta = 1$ implies that

$$\left|x^2-9\right| \le \left|x-3\right|\left(\left|x-3\right|+6\right) < 1(1+6) = 7 < \varepsilon\,.$$

Case 2: If $1 > \frac{\varepsilon}{7}$. Then $\min\{1, \frac{\varepsilon}{7}\} = \frac{\varepsilon}{7}$.

So, $0 < \left|x-3\right| < \delta = \frac{\varepsilon}{7} < 1$ implies that

$$\left|x^2-9\right| \le \left|x-3\right|\left(\left|x-3\right|+6\right) < \tfrac{\varepsilon}{7}(1+6) = \varepsilon\,.$$

Extra example (a)

(a) $\lim_{x\to 2}(x-1)^2 = 1$. Let $\varepsilon > 0$. Set $\delta = \min\left\{1, \frac{1}{3}\varepsilon\right\}$. Then $0 < \left|x-2\right| < \delta$ implies that

$$\left|f(x) - L\right| = \left|(x-1)^2 - 1\right| = \left|x^2 - 2x\right| = \left|x-2\right|\left|x\right| \le \left|x-2\right|\Big(\left|x-2\right| + 2\Big) < \varepsilon\,.$$

Extra example (b)

(b) $\lim\limits_{x\to 1}\left(x^3-x^2+x-1\right)=0$. Note that $x^3-x^2+x-1\equiv(x-1)(x^2+1)$.

Method 1. Set $\delta=\min\left\{1,\frac{\varepsilon}{5}\right\}$. Then $0<\left|x-1\right|<\delta$ implies that

$$\begin{aligned}\left|x^3-x^2+x-1\right| &\equiv \left|(x-1)(x^2+1)\right| \\ &= \left|x-1\right|\left|x^2-1+2\right| \\ &\le \left|x-1\right|\left(\left|x^2-1\right|+2\right) \\ &= \left|x-1\right|\left(\left|x-1\right|\left|x+1\right|+2\right) \\ &\le \left|x-1\right|\left(\left|x-1\right|\left(\left|x-1\right|+2\right)+2\right) \\ &\le \tfrac{\varepsilon}{5}(1(1+2)+2)=\varepsilon\end{aligned}$$

Extra example (b)

Method 2. $\lim_{x\to 1}(x-1)(x^2+1)=0$. If $0<\left|x-1\right|<1$, then

$$x\neq 1 \text{ and } 0<x<2 \Rightarrow 1<x^2+1<5 \Rightarrow \left|x^2+1\right|<5,\ x\neq 1.$$

Set $\delta=\min\left\{1,\frac{\varepsilon}{5}\right\}$. Then $0<\left|x-1\right|<\delta$ implies that

$$\left|x^3-x^2+x-1\right|=\left|x-1\right|\left|x^2+1\right|<\tfrac{\varepsilon}{5}(5)=\varepsilon.$$

Extra example (c)

(c) $\lim\limits_{x\to 1} \frac{x-1}{x+1} = 0\,.$

Choice 1. If $0 < \left|x-1\right| < 1$, then

$$x \neq 1 \text{ and } 1 < x+1 < 3 \Rightarrow \tfrac{1}{3} < \tfrac{1}{x+1} < 1 \Rightarrow \left|\tfrac{1}{x+1}\right| < 1\,,\ x \neq 1\,.$$

Set $\delta = \min\left\{1, \varepsilon\right\}$. Then $0 < \left|x-1\right| < \delta$ implies that $\left|\frac{x-1}{x+1}\right| < \left|x-1\right| < \varepsilon\,.$

Extra example (c)

Choice 2. We can make other choose $\frac{1}{2}$ instead of 1, then the δ will change.

$\lim\limits_{x\to 1}\frac{x-1}{x+1}=0$. If $0<\left|x-1\right|<\frac{1}{2}$, then

$$x\neq 1 \text{ and } \frac{1}{2}<x<\frac{3}{2}\Rightarrow\frac{3}{2}<x+1<\frac{5}{2}\Rightarrow\frac{2}{5}<\frac{1}{x+1}<\frac{2}{3}\Rightarrow\left|\frac{1}{x+1}\right|<\frac{2}{3},\ x\neq 1.$$

Set $\delta=\min\left\{\frac{1}{2},\frac{3\varepsilon}{2}\right\}$. Then $0<\left|x-1\right|<\delta$ implies that

$$\left|\frac{x-1}{x+1}\right|<\frac{2}{3}\left|x-1\right|<\frac{2}{3}\left(\frac{3}{2}\varepsilon\right)=\varepsilon.$$

Extra example (d)

(d) $\lim\limits_{x\to 1}\frac{x+2}{x-2}=-3$. If $0<\left|x-1\right|<\frac{1}{2}$, then $x\neq 1$ and

$$\begin{aligned}
&\tfrac{1}{2}<x<\tfrac{3}{2}\\
&\Rightarrow -\tfrac{3}{2}<x-2<-\tfrac{1}{2}\\
&\Rightarrow \tfrac{1}{2}<2-x=\left|x-2\right|<\tfrac{3}{2}\\
&\Rightarrow \tfrac{2}{3}<\tfrac{1}{\left|x-2\right|}<2 \text{ where } x\neq 1.
\end{aligned}$$

Set $\delta=\min\left\{\frac{1}{2},\frac{\varepsilon}{8}\right\}$. Then $0<\left|x-1\right|<\delta$ implies that

$$\left|\tfrac{x+2}{x-2}+3\right|=4\left|\tfrac{x-1}{x-2}\right|<4\left(2\right)\left(\tfrac{\varepsilon}{8}\right)=\varepsilon\,.$$

Extra example (d)

Remark: If we make the choice $0 < \left|x-1\right| < 1$, then we have

$$-1 < x-1 < 1 \Rightarrow -2 < x-2 < 0\,.$$

This is impossible to take the reciprocal.

Back to Example 6.3

Let $f, g, h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be functions defined by

$$f(x) = \sin\left(\frac{1}{x}\right), \qquad g(x) = x\sin\left(\frac{1}{x}\right), \qquad h(x) = \frac{\sin x}{x}\,.$$

➡ Prove that $\lim\limits_{x\to 0} x\sin\left(\frac{1}{x}\right) = 0\,.$

$$0 < \left|x - 0\right| < \delta \Rightarrow \left|x\sin\left(\frac{1}{x}\right) - 0\right| = \left|x\sin\left(\frac{1}{x}\right)\right| \le \left|x\right| < \delta\,.$$

Choose $\delta = \varepsilon$. Then we have

$$0 < \left|x - 0\right| < \delta \Rightarrow \left|x\sin\left(\frac{1}{x}\right) - 0\right| < \delta = \varepsilon\,.$$

Example

Suppose $\lim_{x \to c} f(x) = L \neq 0$. Prove that $\left|f(x)\right| > \frac{3}{4}\left|L\right|$.

Proof. From the definition,

$\forall \varepsilon > 0 \ \exists \delta > 0$ such that $\left|f(x) - L\right| < \varepsilon$ whenever $0 < \left|x - c\right| < \delta$.

$$\begin{aligned} \text{Now, } \left|L\right| &= \left|L - f(x) + f(x)\right| \\ &\leq \left|L - f(x)\right| + \left|f(x)\right| \\ &< \varepsilon + \left|f(x)\right| \\ \Rightarrow \left|f(x)\right| &> \left|L\right| - \varepsilon \end{aligned}$$

Choose $\varepsilon = \frac{1}{4}\left|L\right| > 0$. Then $\left|f(x)\right| > \frac{3}{4}\left|L\right|$.

This result will be used in proving Theorem 6.5 later.

8. Properties of limits

Theorem 6.5

Let $f, g : D \to \mathbb{R}$ be functions such that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$ for some $a \in \overline{D}$. Let $c \in \mathbb{R}$ be a constant. Then

(a) $\lim_{x \to a} c = c$

(b) $\lim_{x \to a} cf(x) = cL$

(c) $\lim_{x \to a} \Big(f(x) + g(x) \Big) = L + M$

(d) $\lim_{x \to a} \Big(f(x) g(x) \Big) = LM$

(e) $\lim_{x \to a} \Big(f(x) \,/\, g(x) \Big) = L \,/\, M$, provided $g(x) \neq 0$ for all $x \in D$ and $M \neq 0$,

Worksheet 5, Q10

Check out the video 4a for the proof!

Example (Polynomial)

For any polynomial $p(x) = \sum_{k=0}^{n} p_k x^k = p_0 + p_1 x + \cdots + p_n x^n$ of degree $n+1$,

$$\lim_{x \to a} p(x) = p(a)\,.$$

Proof. By the results obtained in the last theorem, i.e., $\lim_{x \to a} c = c$, $\lim_{x \to a} x^k = a^k$, $\lim_{x \to a} p_k x^k = p_k a^k \ \forall k \in \mathbb{N}$. Then

$$\lim_{x \to a} p(x) = \lim_{x \to a} \sum_{k=0}^{n} p_k x^k = \sum_{k=0}^{n} \lim_{x \to a} \left(p_k x^k \right) = \sum_{k=0}^{n} p_k a^k = p(a)\,.$$

Check out the video 4a for the proof!

Theorem

Let $f : D \to \mathbb{R}$ be a function and $c \in \overline{D}$. Suppose $\lim_{x \to c} f(x) = L$.

If $\alpha \boxed{< or \leq} f(x) \boxed{< or \leq} \beta$ for all $x \in D$. Then $\alpha \leq L \leq \beta$.

In other words, if f is bounded, then the limit of f is bounded.

Check out the video 4a for the proof!

Squeeze theorem in function

Let $f, g, h : D \to \mathbb{R}$ be functions and $c \in \bar{D}$.

Suppose $f(x) \leq g(x) \leq h(x)\ \forall x \in D$ and $\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L$.

Then $\lim_{x \to c} g(x) = L$.

The theorem is true when the less than or equal is changed to strictly less than.

Proof. From the definition of limits, $\forall \varepsilon > 0 \ \exists \delta_1, \delta_2 > 0$ such that

$$0 < \left|x - c\right| < \delta_1 \Rightarrow \left|f(x) - L\right| < \varepsilon \Rightarrow L - \varepsilon < f(x) < L + \varepsilon$$
$$0 < \left|x - c\right| < \delta_2 \Rightarrow \left|h(x) - L\right| < \varepsilon \Rightarrow L - \varepsilon < h(x) < L + \varepsilon$$

.

Let $\delta = \min\left\{\delta_1, \delta_2\right\}$. Then

$$L - \varepsilon < f(x) \le g(x) \le h(x) < L + \varepsilon \Rightarrow \left|g(x) - L\right| < \varepsilon$$

whenever $0 < \left|x - c\right| < \delta$.

Theorem 7.2

Theorem 7.2. Sequential Criterion vs Function Limits

Let $f : D \to \mathbb{R}$ be a function and $c \in \bar{D}$.

Let $a : \mathbb{N} \to \mathbb{R}$ be a sequence with $a_n \neq c \ \forall n \in \mathbb{N}$. Then

$$\boxed{\lim_{x \to c} f(x) = L} \Leftrightarrow \boxed{\lim\left(f\left(a_n\right)\right) = L \ \forall \left(a_n\right) \text{ with } \lim\left(a_n\right) = c.}$$

Check out the video 4a for the proof!

Remark

Let $\left(t_n\right), \left(s_n\right)$ be distinct sequences such that

$$\lim_{n\to\infty}\left(t_n\right) = c = \lim_{n\to\infty}\left(s_n\right).$$

$$\text{If } \lim_{x\to c} f(x) = L\,,$$

then $\lim_{n\to\infty} f\left(t_n\right) = L = \lim_{n\to\infty} f\left(s_n\right).$

Corollary 7.3

When apply the contrapositive, we have:

Let $\left(t_n\right), \left(s_n\right)$ be distinct sequences such that

$$\lim_{n\to\infty}\left(t_n\right) = c = \lim_{n\to\infty}\left(s_n\right),$$

but $\lim_{n\to\infty} f\left(t_n\right) \neq \lim_{n\to\infty} f\left(s_n\right)$ (or limit does not exist).

Then $\lim_{x\to c} f(x)$ does not exist.

Back to Example 6.3 again.

Let $f, g, h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be functions defined by

$$f(x) = \sin\left(\frac{1}{x}\right), \qquad g(x) = x\sin\left(\frac{1}{x}\right), \qquad h(x) = \frac{\sin x}{x}.$$

Prove that $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist.

Prove that $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ does not exist.

Proof. Let $f(x)=\sin\left(\frac{1}{x}\right)$. Now, we would like to find two distinct sequences $\left(a_n\right),\left(b_n\right)$ such that $\lim\left(a_n\right)=\lim\left(b_n\right)=0$; but $\lim\left(f(a_n)\right)\neq\lim\left(f(b_n)\right)$. Then $\lim_{x\to a} f(x)$ does not exist. We then choose such a_n and b_n to complete the task.

$$a_n=\frac{1}{\left(2n+\frac{1}{2}\right)\pi}\to 0 \qquad \& \qquad f(a_n)=\sin\left(2n+\tfrac{1}{2}\right)\pi=1 \quad \forall n\in\mathbb{N}$$

$$b_n=\frac{1}{\left(2n-\frac{1}{2}\right)\pi}\to 0 \qquad \& \qquad f(b_n)=\sin\left(2n-\tfrac{1}{2}\right)\pi=-1 \quad \forall n\in\mathbb{N}$$

Hence, $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ does not exist.

9. One-sided limits

Split to left and right

$$0 < |x - c| < \delta$$

means

$$c - \delta < x < c \text{ or } c < x < c + \delta$$

$\swarrow$ $\qquad$ $\searrow$

$c - \delta < x < c$ left limit

$c < x < c + \delta$ right limit

Definitions for one-sided limit

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$.

(i) Then f converges to $L \in \mathbb{R}$ as x tends to c *from the right* if

$$\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \ni \left| f(x) - L \right| < \varepsilon \text{ whenever } c < x < c + \delta \,,\ x \in D \,.}$$

We write this as $\lim\limits_{x \to c^+} f(x) = L$.

(ii) Then f converges to $L \in \mathbb{R}$ as x tends to c *from the left* if

$$\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \ni \left| f(x) - L \right| < \varepsilon \text{ whenever } c - \delta < x < c \,,\ x \in D \,.}$$

We write this as $\lim\limits_{x \to c^-} f(x) = L$.

Comparison of limit definitions

$$\lim_{x \to a} f(x) = L \qquad \exists \delta > 0 \ni 0 < \left| x - a \right| < \delta \Rightarrow \left| f(x) - L \right| < \varepsilon \ \forall \varepsilon > 0 \, .$$

$$\lim_{x \to a^+} f(x) = L \qquad \exists \delta > 0 \ni \boxed{a < x} < a + \delta \Rightarrow \left| f(x) - L \right| < \varepsilon \ \forall \varepsilon > 0 \, .$$

$$\lim_{x \to a^-} f(x) = L \qquad \exists \delta > 0 \ni a - \delta < \boxed{x < a} \Rightarrow \left| f(x) - L \right| < \varepsilon \ \forall \varepsilon > 0 \, .$$

Example 6.7

Let $f : \mathbb{R} \to \mathbb{R}$ be a function given by

$$f(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ -x - 1, & x < 0 \end{cases}.$$

Show that $\lim_{x \to 0^+} f(x) = 0$ and $\lim_{x \to 0^-} f(x) = -1$.

Note that the limit of $f(x)$ when x tends to 0 does not exist..

$$\lim_{x \to 0^+} f(x) = 0$$

Proof. Let $\delta > 0$ and set $\delta = \varepsilon^2$. Then

$$0 < x < \delta \Rightarrow \left|\sqrt{x} - 0\right| = \left|\sqrt{x}\right| = \sqrt{x} < \sqrt{\delta} = \varepsilon\,.$$

$$\lim_{x \to 0^-} f(x) = -1$$

Proof. Let $\delta > 0$ and set $\delta = \varepsilon$. Then

$$-\delta < x < 0 \Rightarrow \left|-x - 1 + 1\right| = \left|-x\right| = -x < \delta = \varepsilon$$

Theorem B

$$\lim_{x \to a} f(x) \text{ exists and equal to } L$$

if and only if

$$\textit{both } \lim_{x \to a^-} f(x) \text{ and } \lim_{x \to a^+} f(x) \text{ exist and equal to } L$$

Remark: If the left and right limits exist but does not equal, then $\lim_{x \to a} f(x)$ does not exist.

Proof

$$\lim_{x \to a} f(x) = L \qquad \exists \delta > 0 \ni 0 < \left| x - a \right| < \delta \Rightarrow \left| f(x) - L \right| < \varepsilon \ \forall \varepsilon > 0 \,. \ (1)$$

$$\lim_{x \to a^+} f(x) = L \qquad \exists \delta > 0 \ni \boxed{a < x} < a + \delta \Rightarrow \left| f(x) - L \right| < \varepsilon \ \forall \varepsilon > 0 \,. \ (2)$$

$$\lim_{x \to a^-} f(x) = L \qquad \exists \delta > 0 \ni a - \delta < \boxed{x < a} \Rightarrow \left| f(x) - L \right| < \varepsilon \ \forall \varepsilon > 0 \,. \ (3)$$

Clearly (2) and (3) implies (1) and (1) implies (2) and (3).

Back to Example 6.3 again.

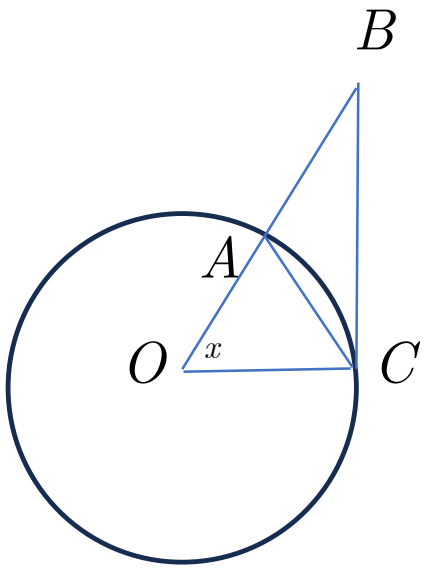
Let $f, g, h : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be functions defined by

$$f(x) = \sin\left(\frac{1}{x}\right), \qquad g(x) = x \sin\left(\frac{1}{x}\right), \qquad \boxed{h(x) = \frac{\sin x}{x}}.$$

Prove that $\lim\limits_{x \to 0} \frac{\sin x}{x} = 1$.

➡ Prove that $\lim\limits_{x\to 0^+}\frac{\sin x}{x}=1$.

Consider a unit circle (radius of 1), let $x=\measuredangle AOC<\frac{\pi}{2}$.



$\text{Area}\,\triangle OAC<\text{Area}\,\overset{\frown}{OAC}<\text{Area}\,\triangle OBC$

Hence, $\frac{1}{2}\sin x<\frac{1}{2}x<\frac{1}{2}\tan x$.

It follows that $1<\frac{x}{\sin x}<\frac{\tan x}{\sin x}=\frac{1}{\cos x}$ or $1>\frac{\sin x}{x}>\cos x$.

Now $\lim\limits_{x\to 0^+}1=1=\lim\limits_{x\to 0^+}\cos x\Rightarrow\lim\limits_{x\to 0^+}\frac{\sin x}{x}=1$ (squeeze theorem).

We can now easily verify that

$\lim_{x \to 0^-} \frac{\sin x}{x} = 1$ by substituting

$u = -x$ to the prior result.

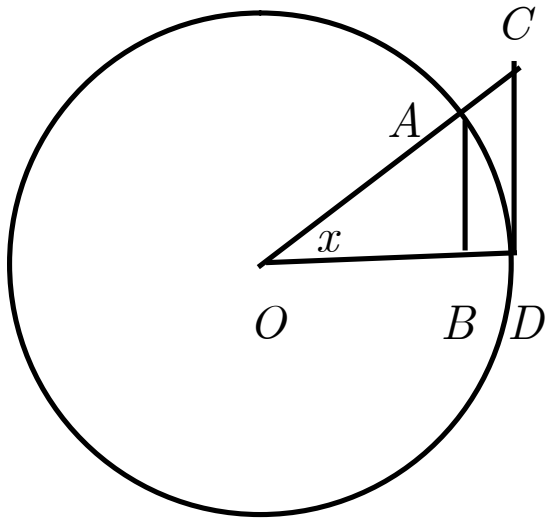
Hence, $\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1}$.

Alternative

Consider a unit circle with origin O, with angle $0 < x < \frac{\pi}{2}$.

Note that the area of $\triangle AOB$ $<$ area of sector $OAD <$ area of $\triangle COD$.

Therefore,



$$\frac{1}{2}OB \times AB < \frac{1}{2}x \times OD^2 < \frac{1}{2}OD \times CD$$

$$\Rightarrow \frac{1}{2}\cos x \sin x < \frac{1}{2}(x)(1)^2 < \frac{1}{2}(\tan x)(1)$$

$$\Rightarrow \cos x \sin x < x < \frac{\sin x}{\cos x}$$

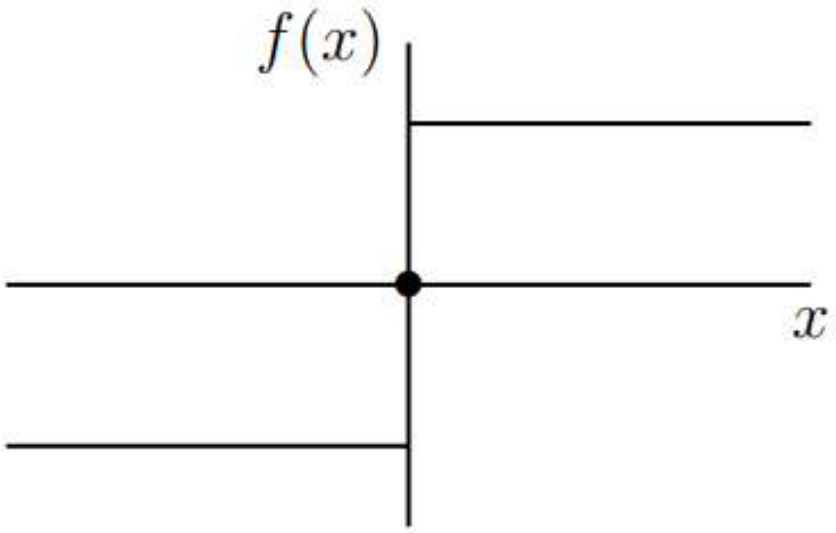
$$\Rightarrow \cos x < \frac{x}{\sin x} < \frac{1}{\cos x}$$

$$\Rightarrow \cos x < \frac{\sin x}{x} < \frac{1}{\cos x} \qquad (\because x \in (0, \tfrac{\pi}{2}) \therefore \sin x, \cos x > 0)$$

Extra example

Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}.$$



It is clear that $\lim_{x \to 0^-} f(x) = -1$ and

$\lim_{x \to 0^+} f(x) = 1$ (prove it!)

However, $\lim_{x \to 0^+} f(x) \neq \lim_{x \to 0^-} f(x)$.

So limit of the function at $x = 0$

does not exist.

How to show a limit (at a point) does not exist?

- Use Corollary 7.3: Find two distinct subsequences that converge to the same limit (at the point), but the function of the subsequences converge differently.
- Use Theorem B: Show that the one-sided (right and left) limits (at the point) does not tally.
- Limit or one-sided limit (at the point) goes to positive/negative infinity.
- Use definition to prove that limit does not exist (by contradiction).

Exercise

Exercise: Find the limit of the following when $x \to 0$, if it exists.

(a) $x \cos\left(\frac{1}{x}\right)$ (b) $\frac{\cos x}{x}$ (c) $\cos\left(\frac{1}{x}\right)$

Can you define these limits?

$$\lim_{\substack{x \to \\ \infty \\ -\infty \\ c \\ c^+ \\ c^-}} f(x) = \begin{matrix} L \\ \infty \\ -\infty \end{matrix}$$